

25F-CY-102
Student ID

Muhammad Rehan
Student Name

Lab 7: Open-Ended Lab-I

Objective:

Students practice basic programming concepts (variables, input/output, operators, conditionals, loops) by designing a small program on their own.

Equipment Needed:

- Computer with a C++ compiler (e.g., g++, Visual Studio)
- Text editor or Integrated Development Environment (IDE)

Lab Task (Open-Ended):

Design and implement a C++ program that solves any simple real-world problem.

The program must take input from the user and display meaningful output.

Requirements (Mandatory):

- Your program must include:
- User input (cin)
- Output (cout)
- At least one condition (if/else)
- At least one loop (for/while)
- Use of variables and operators
- Clear comments

Examples of Problems You Can Choose (Students pick one):

1. **Grade Calculator**
Input marks of 5 subjects → calculate total, percentage, grade.
2. **Simple Calculator**
User enters two numbers and an operator → program outputs result.
3. **Number Analyzer**
Take a number → check if it is even/odd, positive/negative, prime, etc.
4. **Shopping Bill Generator**
Input prices of items → apply tax → print final bill.

5. **Multiplication Table Generator**

User enters a number → print its multiplication table.

6. **Guess the Number Game**

Computer chooses a number → user guesses → give hints (higher/lower).

7. **BMI Calculator**

Input height & weight → compute BMI → categorize.

8. **Temperature Converter**

Convert Celsius ↔ Fahrenheit.

9. **Simple Attendance Tracker**

Input number of classes → input attended classes → compute percentage.

10. **Custom Problem**

Students may invent their own small problem.

Submission Requirements

- Source code file (lab1.cpp)
- Screenshot of program output
- Short report (1 page):
 - o Problem description
 - o Logic flow
 - o Screenshot
 - o Conclusion